

Measurement of Absolute Double Differential Cross-Section in Invariant Mass and p_T Bins

Chatura Kuruppu

SeaQuest Experiment (E906)
Fermi National Accelerator Laboratory

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Note: We do not use RS5a and RS5b files for this study.

ROOT files used:

- merged_RS57_Empty_2.1138_FinalData.root
- merged_RS57_LD2_3.1138_FinalData.root
- merged_RS57_LH2_1.1138_FinalData.root
- merged_RS59_Empty_2.466_FinalData.root
- merged_RS59_LD2_3.466_FinalData.root
- merged_RS59_LH2_1.465_FinalData.root
- merged_RS5a_Empty_2.1680_FinalData.root
- merged_RS5a_LD2_3.1689_FinalData.root
- merged_RS5a_LH2_1.1476_FinalData.root
- merged_RS5b_Empty_2.917_FinalData.root
- merged_RS5b_LD2_3.918_FinalData.root
- merged_RS5b_LH2_1.770_FinalData.root
- merged_RS62_Empty_2.1234_FinalData.root
- merged_RS62_LD2_3.1237_FinalData.root
- merged_RS62_LH2_1.1234_FinalData.root
- merged_RS67_Empty_2.14_FinalData.root
- merged_RS67_LD2_3.15_FinalData.root
- merged_RS67_LH2_1.5_FinalData.root
- merged_RS70_Empty_2.267_FinalData.root
- merged_RS70_LD2_3.266_FinalData.root
- merged_RS70_LH2_1.264_FinalData.root

Kinematic Binning Definition:

- **Mass Bins (GeV):** [4.2, 4.5, 4.8, 5.1, 5.4, 5.7, 6.0, 6.3, 6.6, 6.9, 7.5, 8.7]
- **p_T Bins (GeV):** [0.0, 0.32, 0.49, 0.63, 0.77, 0.95, 1.18, 1.8]
- **x_F Bins:** [0.0, 0.85] with a fixed step width of 0.05.

Event Selection Criteria

Dimuon Kinematics:

- $4.2 < \text{Mass} < 8.8 \text{ GeV}$, $-0.1 < x_F < 0.95$, $0.05 < x_T \leq 0.58$, and $|\cos \theta| < 0.5$.
- Vertex cuts: $-280 < dz < -5$, target/dump tracking separation criteria met.
- Transverse momentum bounds: $|dp_x| < 1.8$, $|dp_y| < 2.0$, and $dp_x^2 + dp_y^2 < 5$.

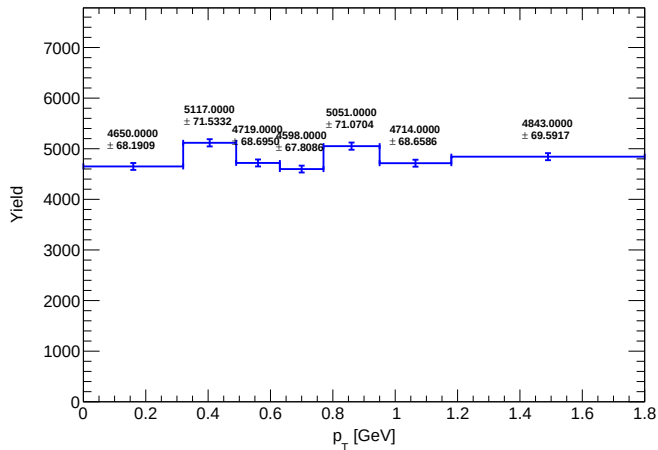
Track Quality & Acceptance:

- $\chi_{\text{target}}^2 < 15$, $\chi_{\text{dimuon}}^2 < 18$, and $\chi^2/(\text{nHits} - 5) < 12$.
- Hit requirements: $\text{nHits} > 13$ per track, $\text{nHits}_1 + \text{nHits}_2 > 29$, $\text{St1 total} > 8$.
- Track vertex longitudinal limits: $-320 < z_v < -5$.
- A dynamic beam offset correction is applied to y-coordinates: offset is 1.6 for $\text{runID} \geq 11000$ and 0.4 otherwise.

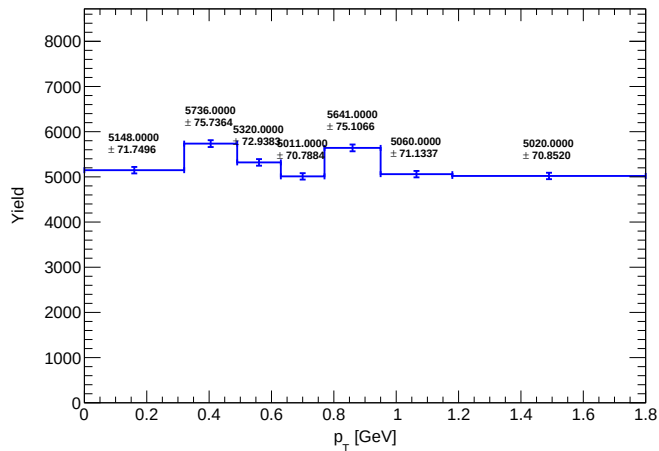
Detector Occupancy Limits:

- Drift chamber hits: $D1 < 400$, $D2 < 400$, $D3 < 400$, and total $D1 + D2 + D3 < 1000$.

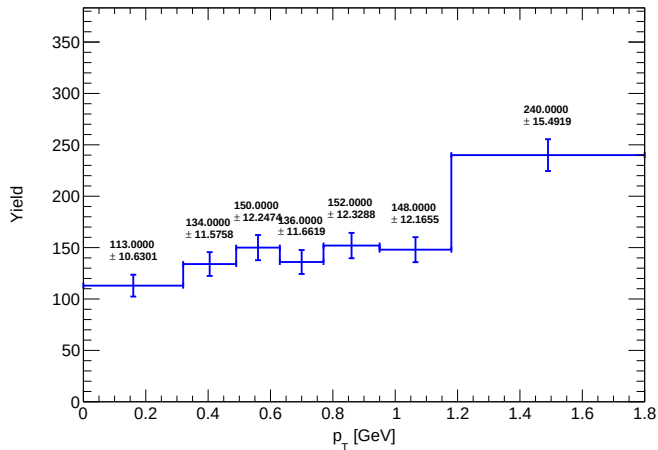
Total Yield (result) (LH2)



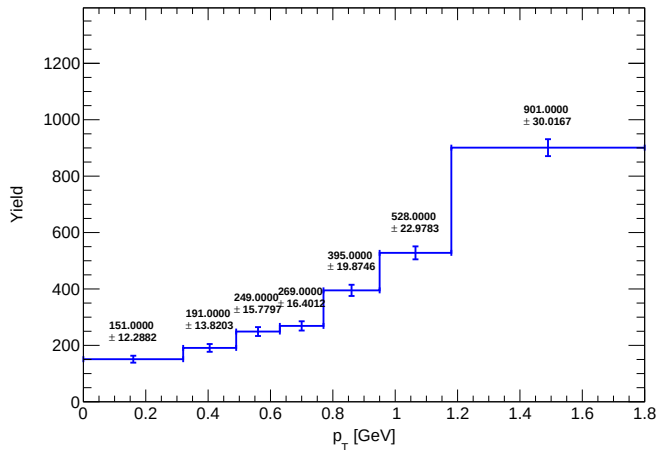
Total Yield (result) (LD2)



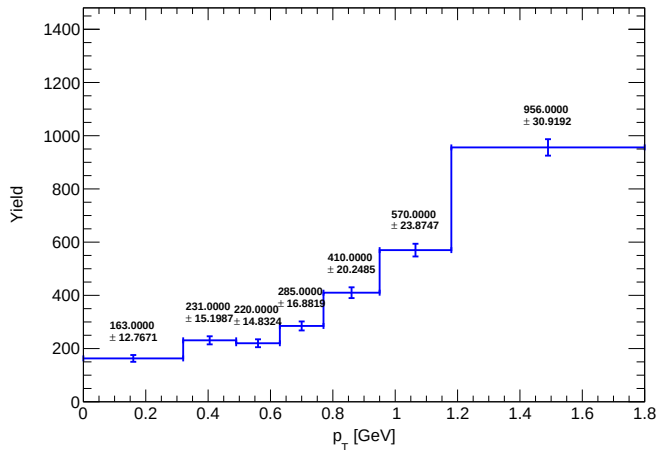
Total Yield (result) (Flask)



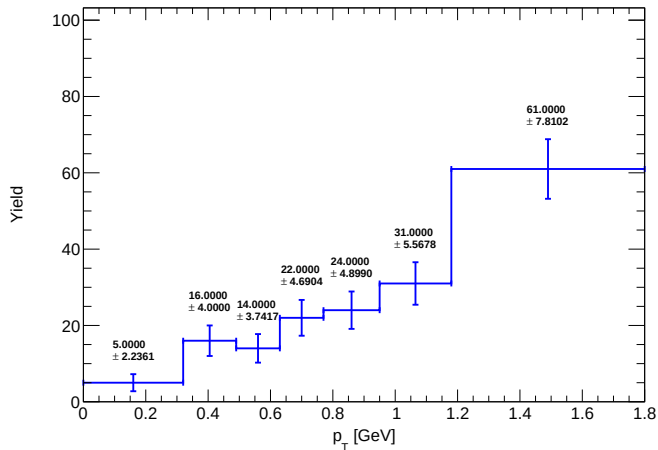
Mix Yield (result_mix) (LH2)



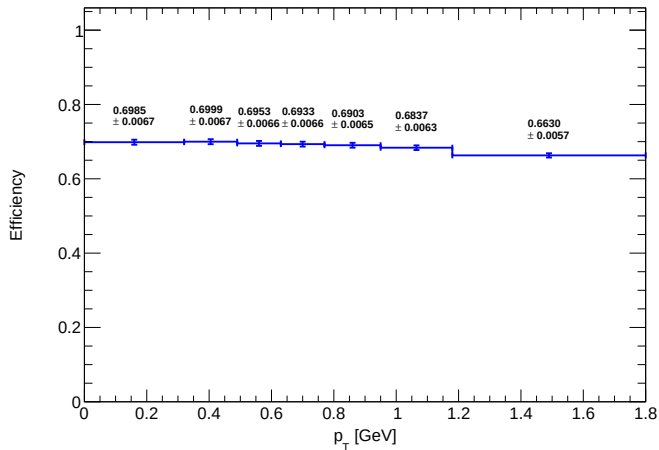
Mix Yield (result_mix) (LD2)



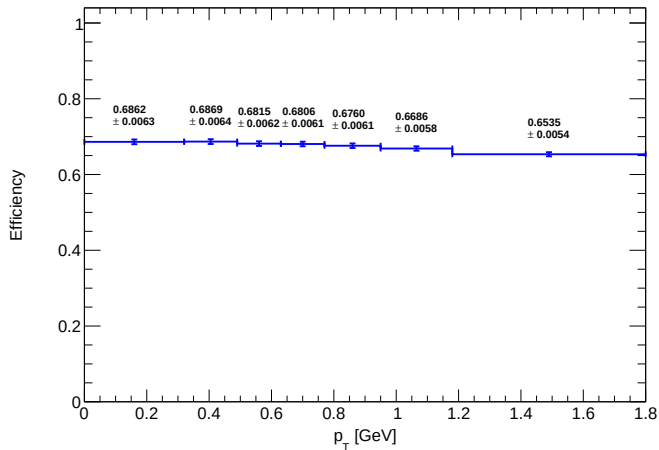
Mix Yield (result_mix) (Flask)



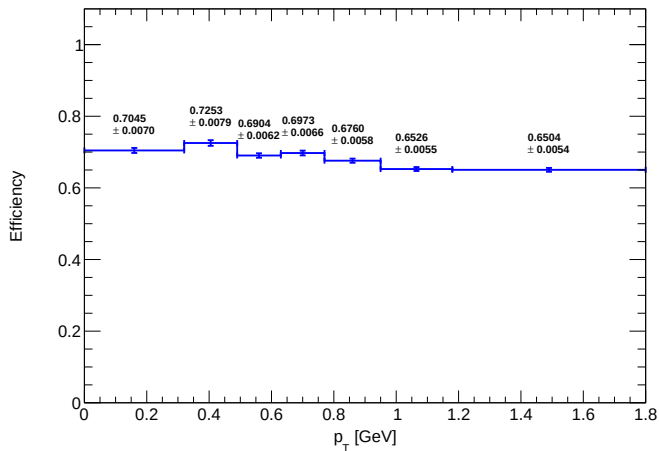
Avg Reco Eff (Total) (LH2)



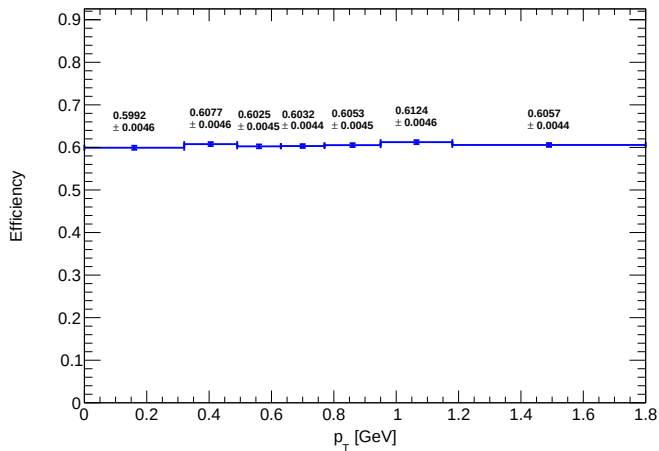
Avg Reco Eff (Total) (LD2)



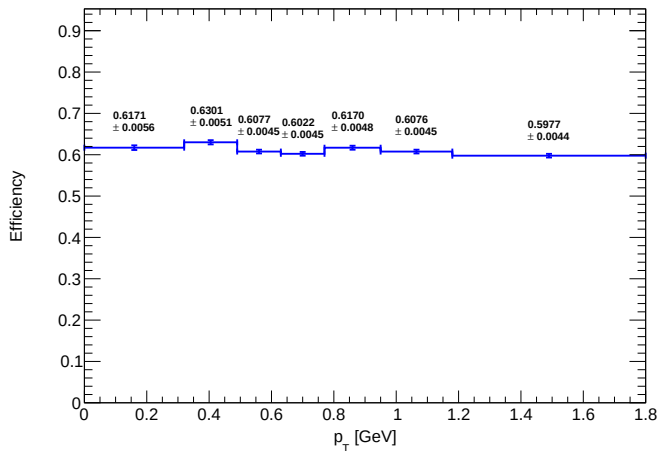
Avg Reco Eff (Total) (Flask)



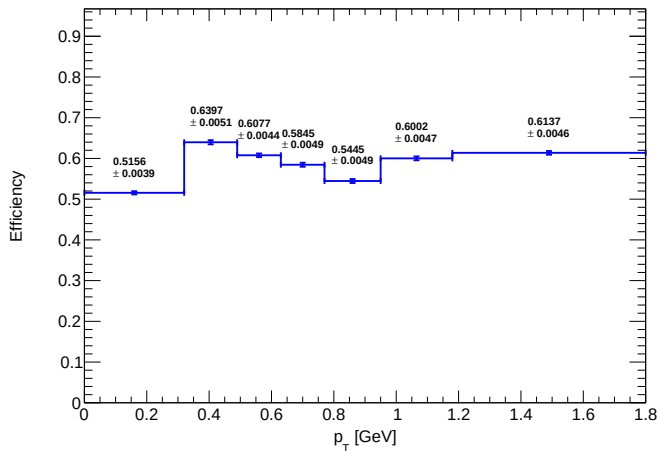
Avg Reco Eff (Mix) (LH2)



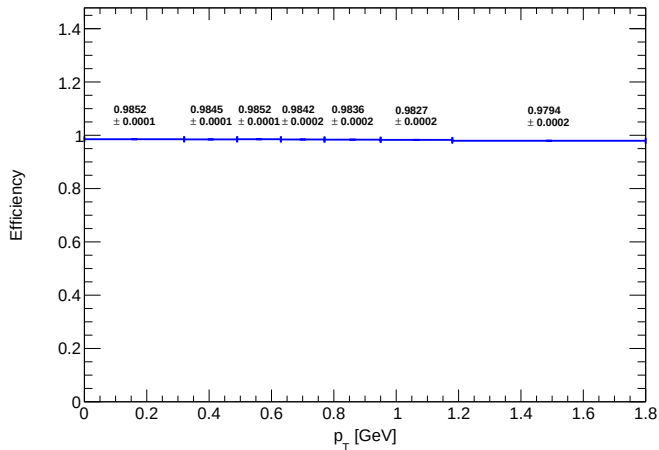
Avg Reco Eff (Mix) (LD2)



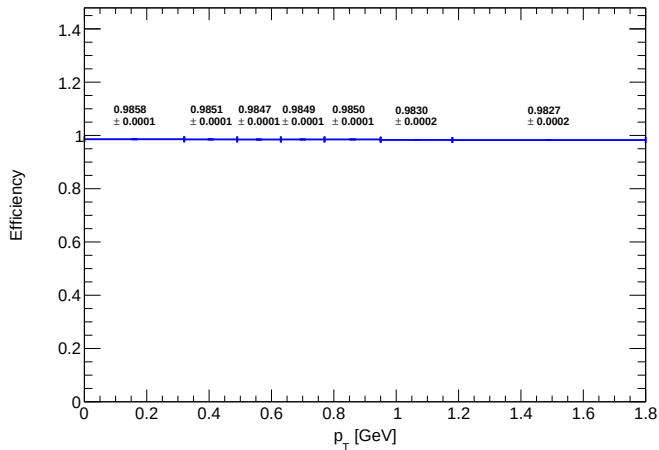
Avg Reco Eff (Mix) (Flask)



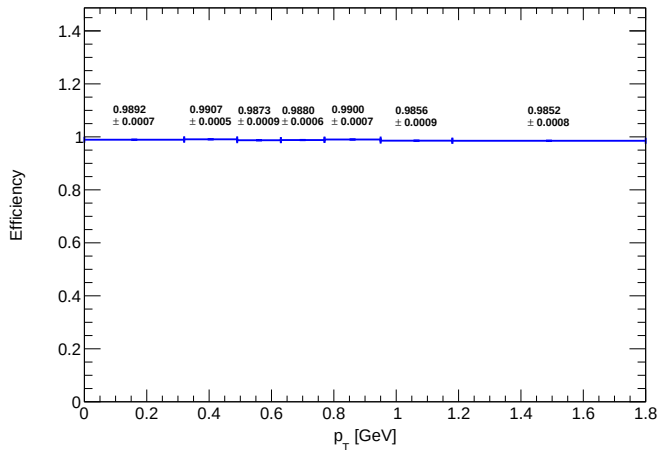
Avg Hodo Eff (Total) (LH2)



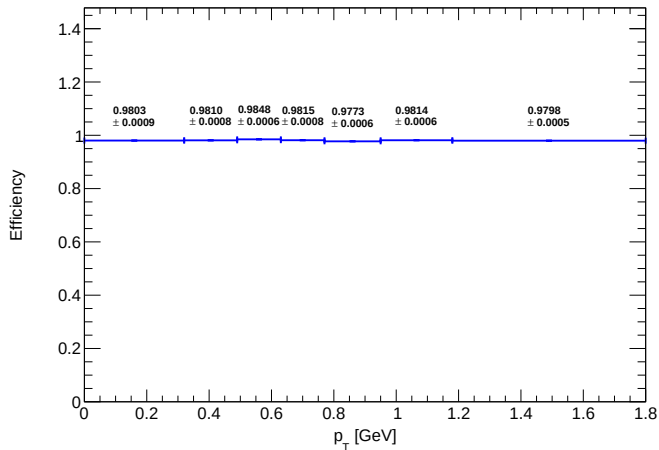
Avg Hodo Eff (Total) (LD2)



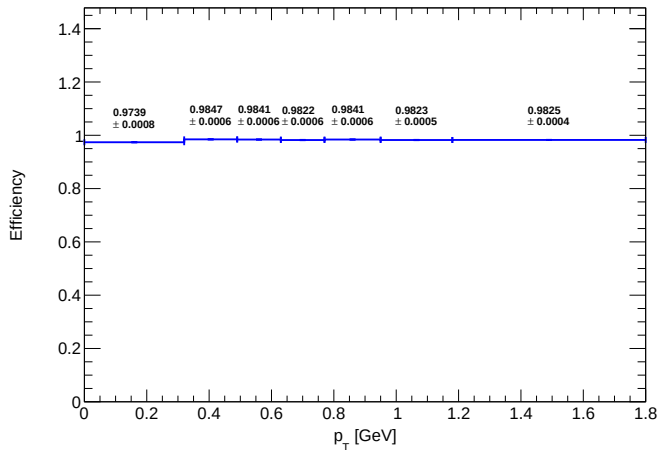
Avg Hodo Eff (Total) (Flask)



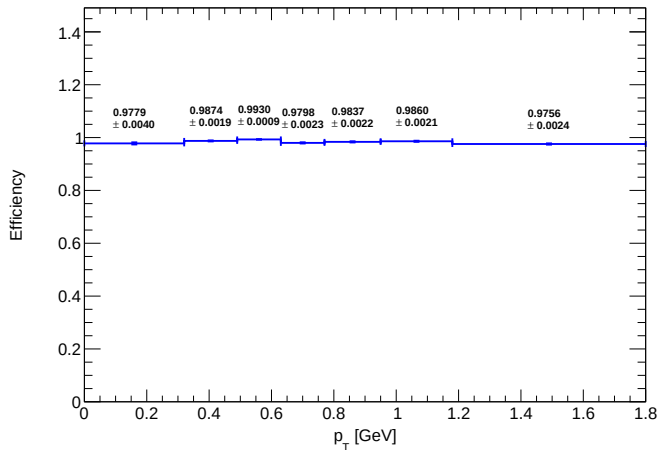
Avg Hodo Eff (Mix) (LH2)



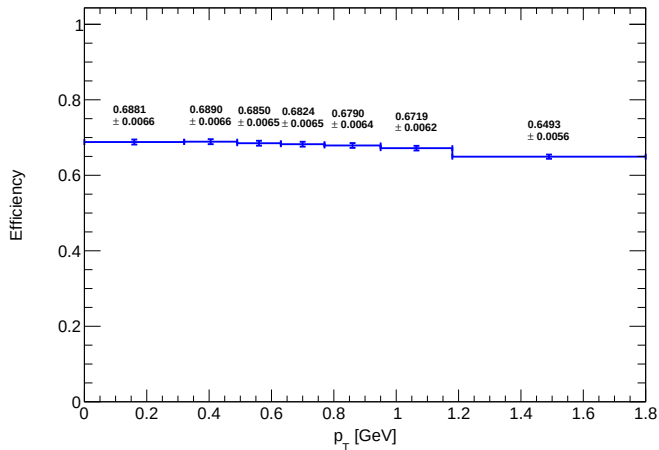
Avg Hodo Eff (Mix) (LD2)



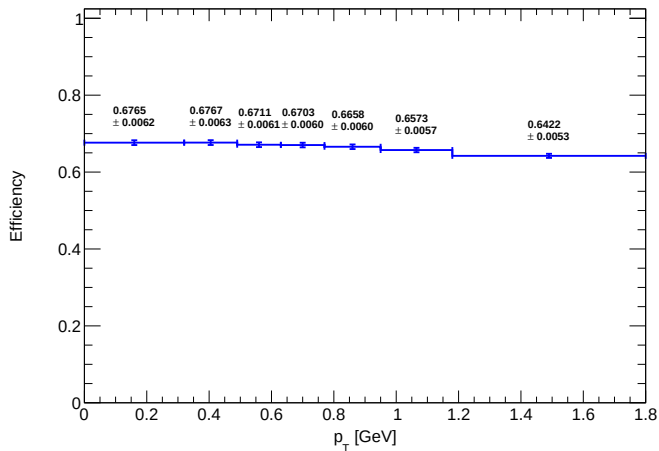
Avg Hodo Eff (Mix) (Flask)



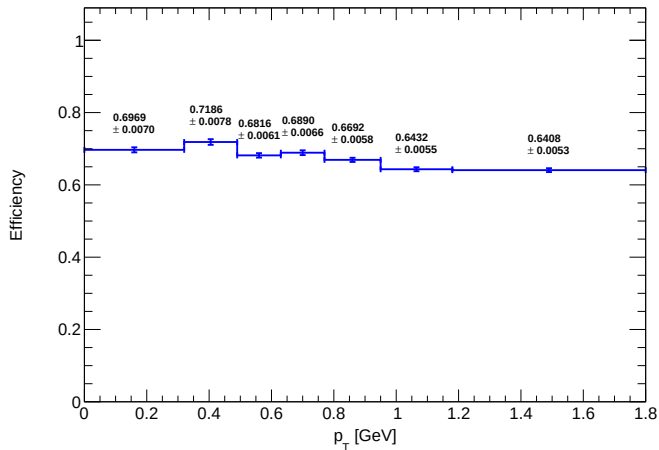
Avg Final Eff (Total) (LH2)

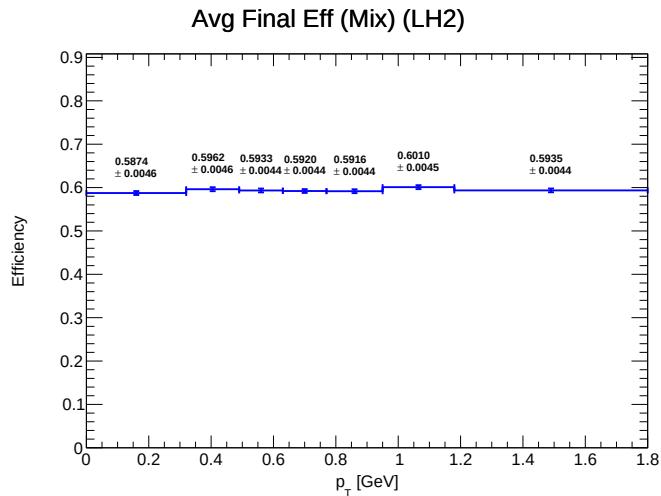


Avg Final Eff (Total) (LD2)

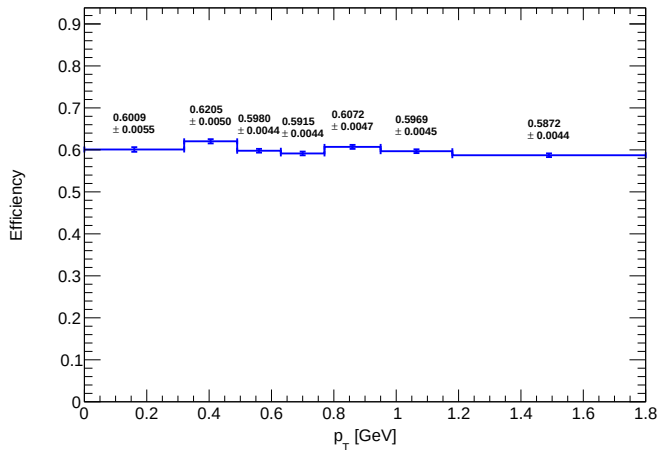


Avg Final Eff (Total) (Flask)

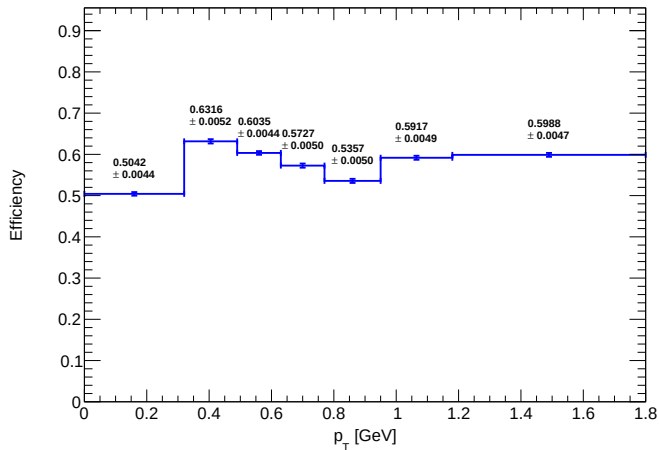




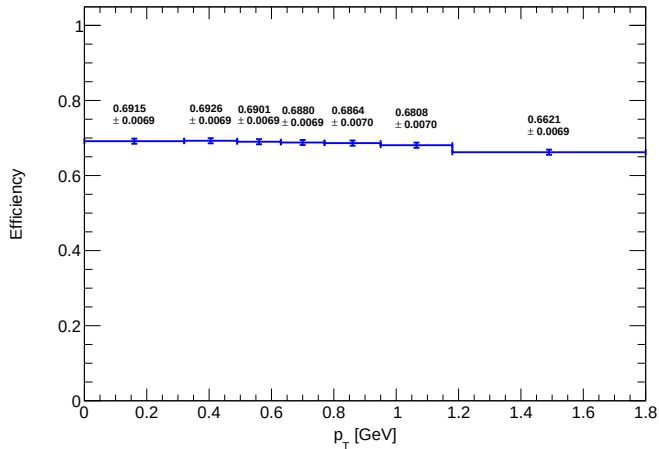
Avg Final Eff (Mix) (LD2)



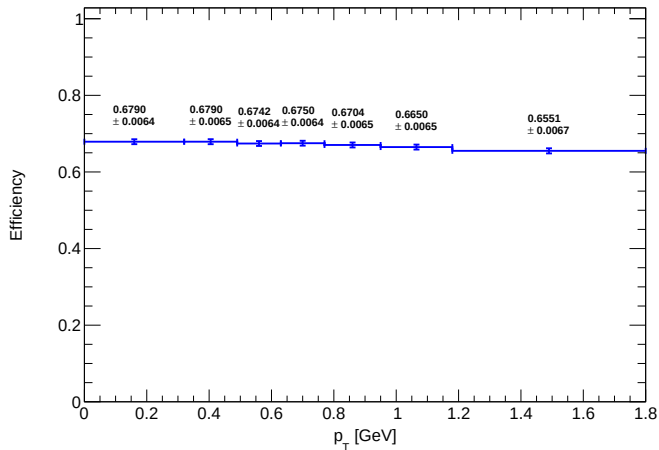
Avg Final Eff (Mix) (Flask)



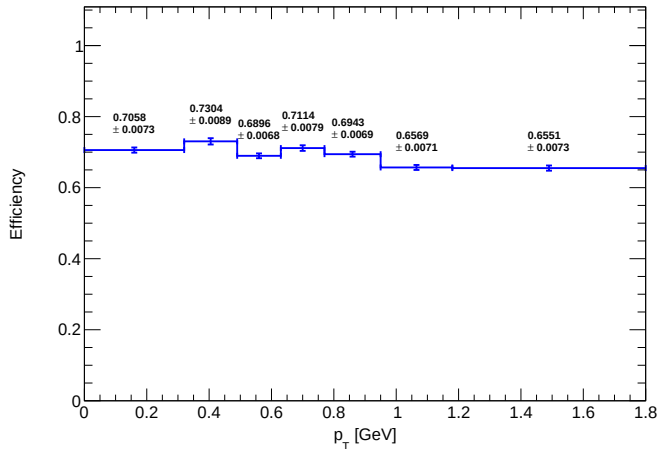
Avg Signal Efficiency (LH2)



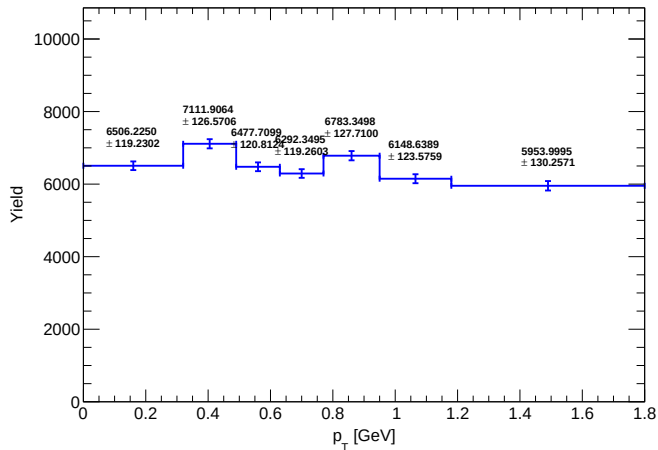
Avg Signal Efficiency (LD2)



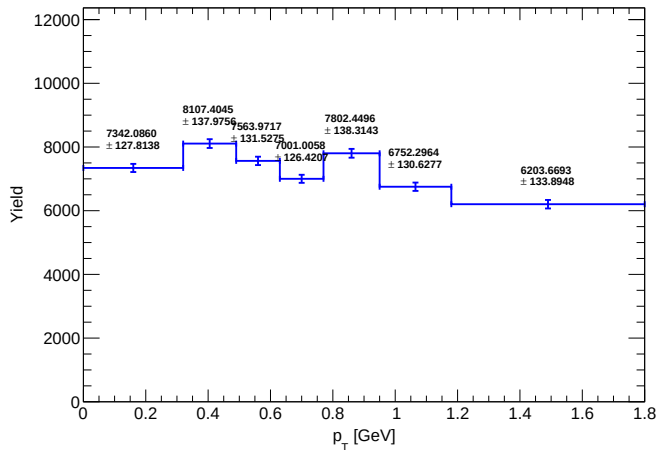
Avg Signal Efficiency (Flask)



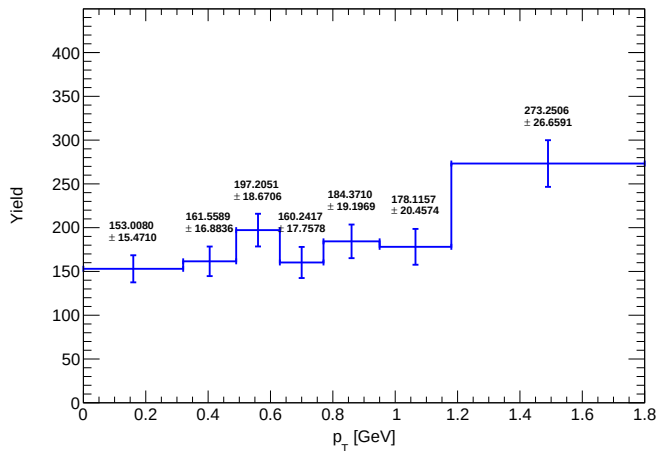
Corrected Yield (Total Error) (LH2)



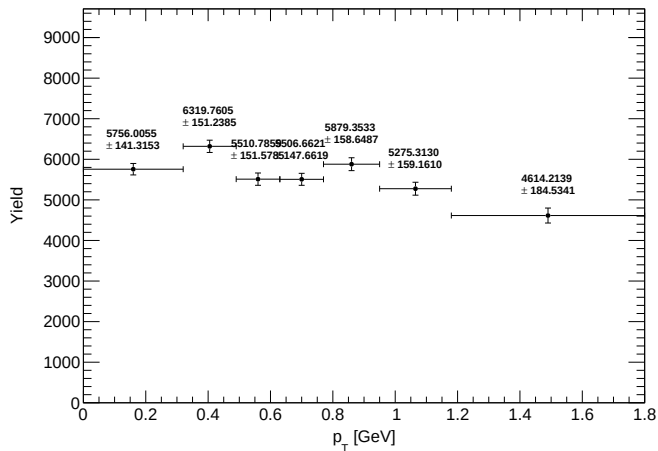
Corrected Yield (Total Error) (LD2)



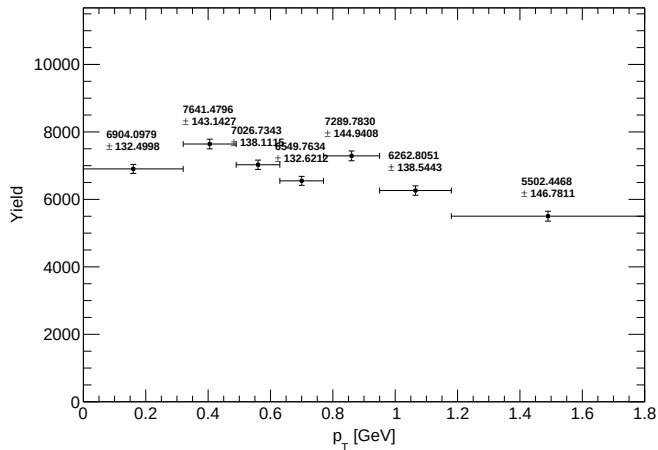
Corrected Yield (Total Error) (Flask)



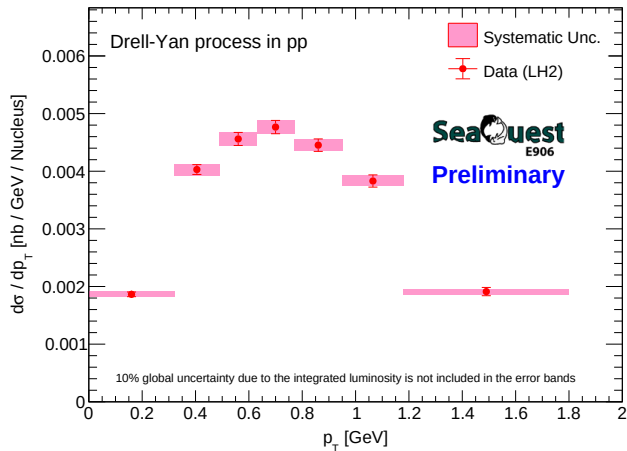
Corrected Yield (LH2 - Flask)



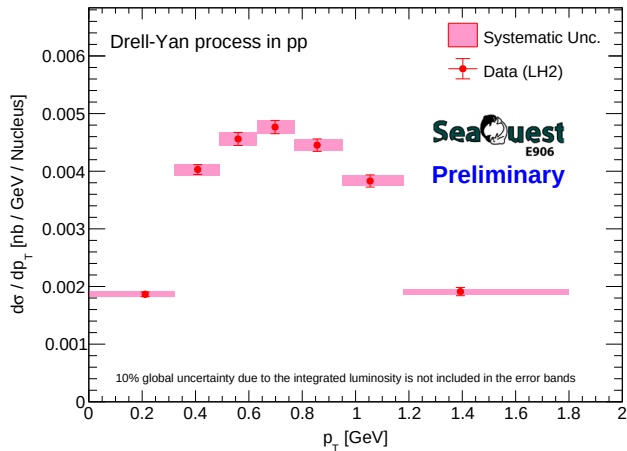
Corrected Yield (LD2 - LH2 - Flask)



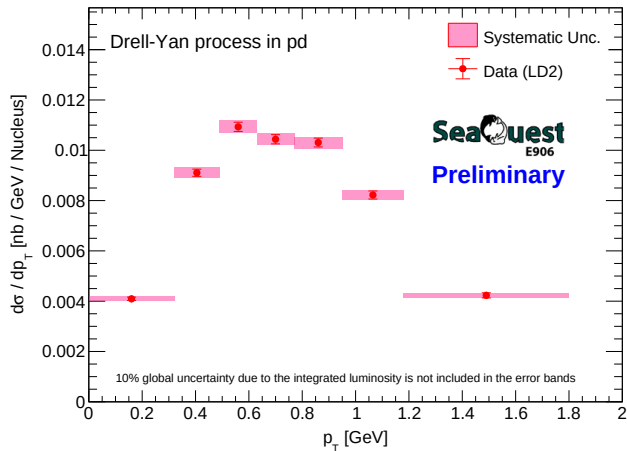
CrossSection_LH2_geom_vs_pT_with_logo



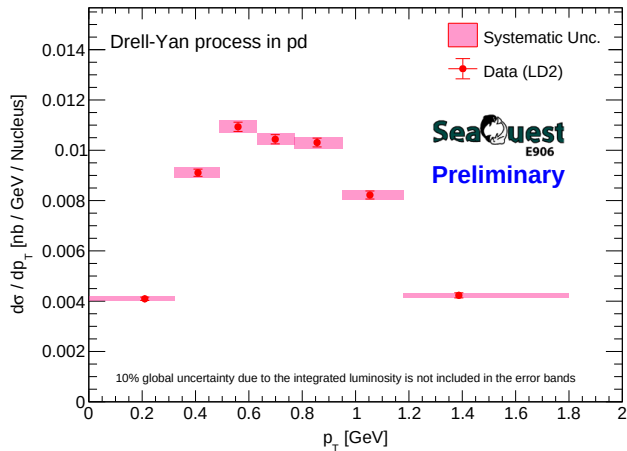
CrossSection_LH2_true_pt_vs_pT_with_log



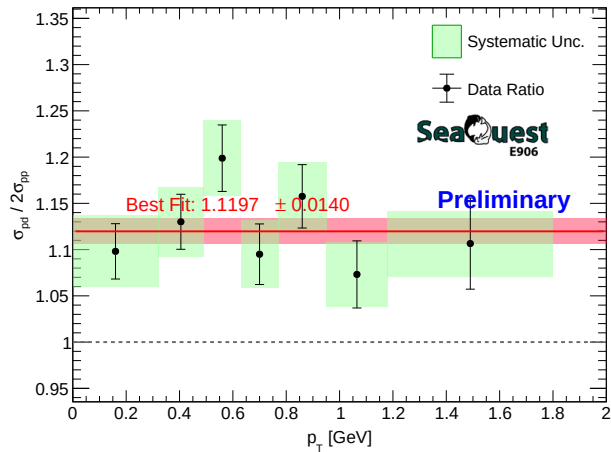
CrossSection_LD2_geom_vs_pT_with_logo



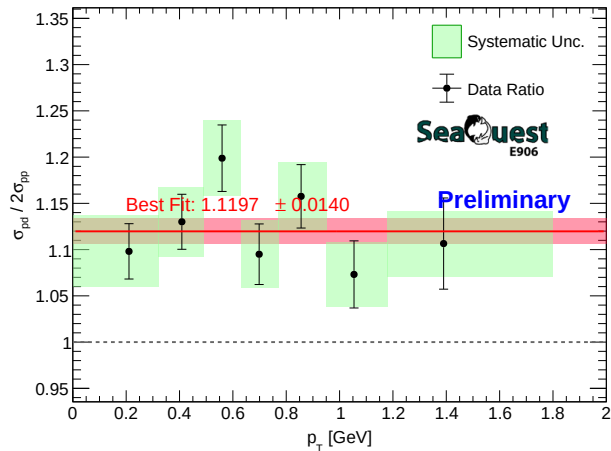
CrossSection_LD2_true_pt_vs_pT_with_log



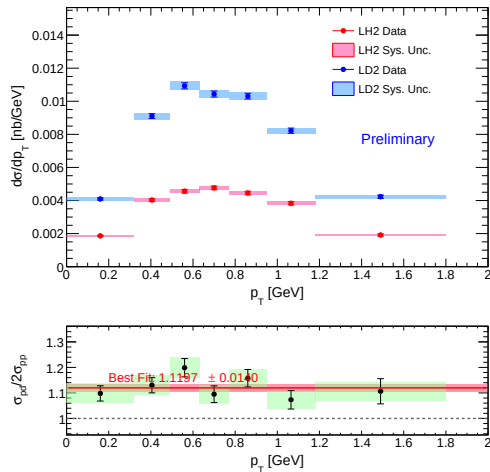
CrossSection_Ratio_pd_2pp_vs_pT_geom_with_logo



CrossSection_Ratio_pd_2pp_vs_pT_true_pt_with_logo



Combined_XSec_Ratio_vs_pT_geom



Combined_XSec_Ratio_vs_pT_true_pt

